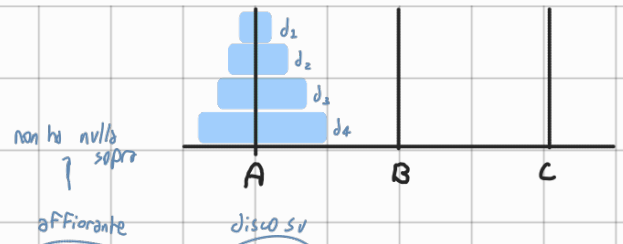


Hanoi PDDL



$$\mathcal{P} = \{ \text{Asse}/1, \text{Disco}/1, \text{smaller}/2, \text{Free}/1, \text{diskOn}/2 \}$$

$$\mathcal{F} = \{ d_1, d_2, d_3, d_4, A, B, C \}$$

$$S_0 = \text{Asse}(A) \wedge \text{Asse}(B) \wedge \text{Asse}(C) \wedge$$

$$\text{Disco}(d_1) \wedge \text{Disco}(d_2) \wedge \text{Disco}(d_3) \wedge \text{Disco}(d_4) \wedge$$

$$\text{smaller}(d_1, d_2) \wedge \text{smaller}(d_1, d_3) \wedge \text{smaller}(d_1, d_4) \wedge$$

$$\text{smaller}(d_2, d_3) \wedge \text{smaller}(d_2, d_4) \wedge \text{smaller}(d_3, d_4) \wedge \text{Free}(B) \wedge \text{Free}(C)$$

$$\text{Free}(d_1) \wedge \text{diskOn}(d_4, A) \wedge \text{diskOn}(d_3, d_4) \wedge \text{diskOn}(d_2, d_3) \wedge \text{diskOn}(d_1, d_2)$$

$$\wedge \text{smaller}(d_1, A) \wedge \text{smaller}(d_2, B) \dots$$

$$\text{Sposta}(X, Y', Y)$$

-Pre

$$\text{Disco}(X) \wedge \text{Free}(X) \wedge \text{Free}(Y) \wedge \text{smaller}(X, Y)$$

$$\text{diskOn}(X, Y') \wedge Y \neq Y'$$

si: diskOn in τ Funz

$$\left(\text{Disk}(Y) \wedge \text{smaller}(X, Y) \right) \vee \text{Asse}(Y)$$

-effetto

$$\neg \text{Free}(Y) \wedge \text{diskOn}(X, Y) \wedge \neg \text{diskOn}(X, Y')$$

State obiettivo non ha la closed-world assumption

Obiettivo: $\text{diskOn}(d_4, B) \wedge \text{diskOn}(d_3, d_4) \wedge \text{diskOn}(d_2, d_3) \wedge \text{diskOn}(d_1, d_2)$